ELEMENTS LIFTING A BASIS OF THE CONORMAL SPACE OF A REGULAR IDEAL GENERATE A REGULAR SEQUENCE LOCALLY

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 4716-4734 and proof span 4736-4767 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof's three parts share one shrinking argument, and the author's remark that the assumptions of the later parts are stronger is preserved.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A quasi-regular sheaf of ideals is locally generated by a quasi-regular sequence, and then its conormal sheaf is free over the quotient with rank the length of that sequence. Cited at proof lines 4736-4742. Produces the reference sequence and shows that the number of given elements equals its length.
- The Koszul complex of a sequence changes by an invertible change of basis without changing its homology, cited from the more-algebra chapter. Cited at proof lines 4763-4766. Transfers H1-regularity and Koszul-regularity from the reference sequence to the given one.
- Invertibility of a matrix over a local ring is detected modulo the maximal ideal, so a matrix invertible at a point is invertible on a neighbourhood. Used at proof lines 4747-4752 and again at 4759-4762. Lets the change of basis be inverted after shrinking, and gives that the given elements form a basis of the conormal sheaf locally.

Source reference. Original lemma 063H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7831–7831. Same-chapter reference used in the retained author text at source line(s) 4742, 5013. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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Source reference. Original lemma 0625: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7784–7784. Same-chapter reference used in the retained author text at source line(s) 4764. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a locally ringed space. Let $\mathcal{J} \subset \mathcal{O}_X$ be a sheaf of ideals. Let $x \in X$ and $f_1, \ldots, f_r \in \mathcal{J}_x$ whose images give a basis for the $\kappa(x)$-vector space $\mathcal{J}_x/\mathfrak{m}_x\mathcal{J}_x$.*

1. *If $\mathcal{J}$ is quasi-regular, then there exists an open neighbourhood such that $f_1, \ldots, f_r \in \mathcal{O}_X(U)$ form a quasi-regular sequence generating $\mathcal{J}|_U$.*
2. *If $\mathcal{J}$ is H_1-regular, then there exists an open neighbourhood such that $f_1, \ldots, f_r \in \mathcal{O}_X(U)$ form an H_1-regular sequence generating $\mathcal{J}|_U$.*
3. *If $\mathcal{J}$ is Koszul-regular, then there exists an open neighbourhood such that $f_1, \ldots, f_r \in \mathcal{O}_X(U)$ form an Koszul-regular sequence generating $\mathcal{J}|_U$.*

Proof. First assume that $\mathcal{J}$ is quasi-regular. We may choose an open neighbourhood $U \subset X$ of x and a quasi-regular sequence $g_1, \ldots, g_s \in \mathcal{O}_X(U)$ which generates $\mathcal{J}|_U$. Note that this implies that $\mathcal{J}/\mathcal{J}^2$ is free of rank s over $\mathcal{O}_U/\mathcal{J}|_U$ (see Lemma 063H and its proof) and hence $r = s$. We may shrink U and assume $f_1, \ldots, f_r \in \mathcal{J}(U)$. Thus we may write

$$f_i = \sum a_{ij} g_j$$

for some $a_{ij} \in \mathcal{O}_X(U)$. By assumption the matrix $A = (a_{ij})$ maps to an invertible matrix over $\kappa(x)$. Hence, after shrinking U once more, we may assume that (a_{ij}) is invertible. Thus we see that $f_1, \ldots, f_r$ give a basis for $(\mathcal{J}/\mathcal{J}^2)|_U$ which proves that $f_1, \ldots, f_r$ is a quasi-regular sequence over U.

Note that in order to prove (2) and (3) we may, because the assumptions of (2) and (3) are stronger than the assumption in (1), already assume that $f_1, \ldots, f_r \in \mathcal{J}(U)$ and $f_i = \sum a_{ij} g_j$ with (a_{ij}) invertible as above, where now $g_1, \ldots, g_r$ is a H_1-regular or Koszul-regular sequence. Since the Koszul complex on $f_1, \ldots, f_r$ is isomorphic to the Koszul complex on $g_1, \ldots, g_r$ via the matrix (a_{ij}) (see More on Algebra, Lemma 0625) we conclude that $f_1, \ldots, f_r$ is H_1-regular or Koszul-regular as desired. □